

Development of Pre-clinical System for Shaping Plasticity via Reflex Conditioning paired with Vagus Nerve Stimulation

Addressing Feedback

Dr. Eric Meyers:

Specific Aims: Development of Pre-clinical System for Shaping Plasticity via Reflex Conditioning paired with Vagus Nerve Stimulation

Interventions that foster neural plasticity in spared sensorimotor networks have long been recognized for their potential to restore functional after neurological injury. However, meaningful recovery from severe injuries necessitates an extensive and precise reorganization of neural connections [cite]. While various methods have been proposed to enhance synaptic plasticity to facilitate this “rewiring,” many have struggled to transition to successful clinical applications. These historical failures are often attributed to the significant challenge of precisely targeting specific, task-relevant synapses and lack of testing in models that reflect clinical environments. [Need to cite this whole section.]

Vagus nerve stimulation (VNS) has surfaced as a potent neuromodulatory strategy to accelerate recovery following conditions like stroke or spinal cord injury. VNS works by triggering the widespread, phasic release of key neuromodulators—specifically acetylcholine, norepinephrine, and serotonin—which collectively prime the central nervous system environment for plasticity. While this global release can facilitate long-lasting improvements in motor function, VNS therapy may be limited when administered without a way to focus these effects, hinting at a lack of specificity of VNS therapy alone. This is because VNS-directed plasticity utilizes “functional targeting” to reinforce only the specific neural networks active during stimulation. Since global neuromodulator release alone lacks precision, VNS must be paired with targeted, circuit-oriented interventions to address the maladaptive hyperreflexia that causes spasticity. This combination allows VNS to effectively treat the underlying neural causes of specific motor dysfunctions. [Need to cite this whole section]

Commented [MEC1]: Think about how this is setting up your specific problem and solution. Not sure where this is headed exactly

Commented [MEC2]: be more specific. Look at recent reviews from Dr Hays that can show you how to describe this

Commented [MEC3]: We’ve shown this in multiple studies, so you can be more direct. VNS does not support plasticity when unpaired.

Commented [MEC4]: Look at Seth’s recent review to see how to describe this.

Commented [MEC5]: Be more specific

This project aims to address that gap by supplementing a VNS-based therapy paradigm with an intervention known as H-Reflex Operant Conditioning (HROC), a targeted therapy known to drive plasticity to the spinal cord, to form a new therapy, VNS-enhanced H-reflex Operant Conditioning (VeHROC). This paradigm will necessitate the development of a pre-clinical system capable of capturing electrophysiological locomotor data from the lower limb muscles and delivering precisely timed stimulations of VNS to enhance the therapeutic effects of HROC. Both effects of HROC and implanted VNS will be evaluated.

By comparing HROC and VeHROC with identical investigative pipelines, I aim to determine whether this combination/addition of VNS with HROC will augment spinal conditioning effects and lead to superior locomotor recovery compared to either therapy utilized alone.

This research will focus on developing multi-modal pre-clinical system capable of real-time recording and deliver VNS and evoke the Hoffman Reflex (H-Reflex) to study the progression of locomotor recovery with electrophysiology metrics such as electromyography (EMG), electroencephalography (EEG), and diaphragm electromyography (dEMG). Ultimately, these efforts support the advancement of rehabilitative therapies that address the underlying spinal circuits of motor disorders and advancing translation toward clinical therapies targeted at driving plasticity to the spinal cord.

Impact Statement:

Aim 1: Develop and Validate Pre-Clinical Technology for Real-Time Recording and Stimulation

Build a closed-loop lab-based system that can simultaneously record biological signals and deliver precisely timed stimulation in response to designed behavioral task paradigms. This system will acquire EMG to measure muscle activity in multiple anatomical locations (leg, jaw, diaphragm), as well as EEG to monitor brain activity and network synchrony. This system will also be equipped to deliver stimulation to different nerves, namely the tibial nerve to evoke the H-Reflex (a measure of spinal reflex excitability) and the left vagal nerve for Vagus Nerve Stimulation (VNS). This system will include an Open Ephys Acquisition Board for acquiring data, as well as programmable 4100 AM Stimulators for the main pieces of instrumentation. Additional components will include a commutator to allow communication between the instrumentation and the rat cage/booth. Recording hardware will include custom built multi-channel headcaps and recording headstages, recording electrodes (EMG and EEG), and lastly active components will include nerve stimulating cuffs (VNS Cuffs and Tibial Cuffs).

The system will be built alongside an in-house developed python software, The H-Reflex App, that will interface mainly with the Open Ephys GUI & Open Ephys Acquisition Board, as well as the programmable 4100 AM Stimulator(s). The H-Reflex App will allow for autonomous execution of experiments including recording electrophysiological activity, eliciting stimulation events, and delivering reward stimulations on successful trials when necessary. The system will be configured with TTL sync lines to enable time-synchronized recording and real-time event triggering. This system will serve as the multi-faceted platform for experimental-behavioral paradigms to support all aspects of the project.

Aim 2: Observe Long-term Electrophysiological Effects of H-Reflex Conditioning in Awake Rats

Evaluate electrophysiological signals and method of collection

Aim 3: Identify Reliable Electrophysiological Biomarkers of VNS for Closed-Loop VNS Delivery

Compare ...

Commented [MEC6]: You introduce HROC out of nowhere. Spend more time above motivating it. I would probably approach it by first describing the clinical need (spasticity), and how we currently don't have any tools to target spasticity with VNS

Commented [MEC7]: Think about how to more cohesively tie in the biomarker piece.

Just off the top of my head, I would set up your introduction like this:

- clinical problem (stroke, SCI, etc.)
- VNS has shown promise for improving motor function in these disorders
- However, there are two key limitations
 - 1) we don't currently have technology to target spasticity in patients
 - 2) We have no way of optimizing VNS effects in patients, so some people get ineffective dosing parameters

Then that sets up both of your key problems

Stages of H-reflex App

S1: Recruitment Curve

S2: Control-Mode

|

→ S3: Down-conditioning with Pellet

→ S4: Up-conditioning with Pellet

→ S5: Down-conditioning with VNS

→ S6: Up-conditioning with VNS

Trial per-Trial for Conditioning

Run S2: (To get historical data and for setting initial threshold)

Run S3/S4/S5/S6: Depending on goal

S3-6 will have a moving window of 100 trials (1 in, 1 out), and everything recalculate the distribution and threshold for the 30th percentile

APPROACH

Aim 1:

Methods

Pre-Clinical Studies

Study Design and Recordings

Cohort

Data Analysis

Expected Results and Interpretations

Potential Pitfalls, Contingencies, and Alternative Approaches

Summary of Aim 1

Aim 2:

Methods

Pre-Clinical Studies

Study Design

Participants

Data Analysis

Expected Results and Interpretations

Potential Pitfalls, Contingencies, and Alternative Approaches

Summary of Aim 2

Aim 3:

Methods

Pre-Clinical Studies

Study Design

Participants

Data Analysis

Expected Results and Interpretations

Potential Pitfalls, Contingencies, and Alternative Approaches

Summary of Aim 3

Apparatus and Procedures

IACUC Standards and Guidelines

Rationale and Significance

The Need for Biomarker-Guided Rehabilitation

Spinal cord injury and stroke are ...

Importance of Real-Time Multimodal Platforms

Vagus nerve stimulation (VNS) ...

Relevance of Comparing Implanted VNS and tAN

Longitudinal Biomarkers as Predictors of Recovery

Broader Impact and Future Applications

Project Timeline

I expect the remaining proposed aims will take two years to complete.

Task	Year 1	Year 2	Year 3	Year 4
Aim 1				
Aim 2				

Aim 3				
Manuscript Prep				

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Citations (From Austin's Document):

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